

PROJECT 3 – ECOMMERCE CUSTOMER RETENTION & CHURN ANALYSIS

DAX MEASURES USED

Revenue Metrics

Total Revenue

```
CALCULATE (
    SUM(ecommerce_orders[revenue]),
    ecommerce_orders[order_status] = "Delivered"
)
```

Purpose: Total revenue from delivered orders.

Total Orders

```
CALCULATE (
    DISTINCTCOUNT(ecommerce_orders[order_id]),
    ecommerce_orders[order_status] = "Delivered"
)
```

Purpose: Total completed orders.

Average Order Value (AOV)

```
DIVIDE([Total Revenue],[Total Orders],0)
```

Purpose: Average revenue per order.

Cancellation Rate

```
DIVIDE (
    CALCULATE (
        COUNT(ecommerce_orders[order_id]),
        ecommerce_orders[order_status]="Cancelled"
    )
)
```

```
),
COUNT(ecommerce_orders[order_id]),
0
)*100
```

Purpose: Percentage of cancelled orders.

Customer Analytics

Total Customers

```
DISTINCTCOUNT(ecommerce_customers[customer_id])
```

Purpose: Total unique customers.

Returning Customers

```
CALCULATE (
    DISTINCTCOUNT(ecommerce_orders[customer_id]),
    FILTER(
        VALUES(ecommerce_orders[customer_id]),
        CALCULATE (
            DISTINCTCOUNT(ecommerce_orders[order_id])
        ) > 1
    )
)
```

Purpose: Customers with multiple purchases.

ARPU (Average Revenue Per User)

```
DIVIDE([Total Revenue],[Total Customers],0)
```

Purpose: Average revenue generated per customer.

VIP Revenue %

```
DIVIDE (
```

```
CALCULATE (
    SUM(ecommerce_orders[revenue]),
    ecommerce_customers[customer_segment]="VIP"
),
[Total Revenue],
0
)
```

Purpose: Revenue contribution from VIP customers.

Retention & Churn Metrics

Repeat Purchase Rate

```
DIVIDE (
    [Returning Customers],
    [Total Customers],
    0
) * 100
```

Purpose: Measures customer loyalty.

Retention Rate

```
DIVIDE (
    [Returning Customers],
    [Total Customers],
    0
) * 100
```

Purpose: Percentage of retained customers.

Churned Customers

```
CALCULATE (
    DISTINCTCOUNT(ecommerce_orders[customer_id]),
    FILTER (
        VALUES(ecommerce_orders[customer_id]),
        CALCULATE (
```

```

        DATEDIFF (
            MAX (ecommerce_orders[order_date]),
            DATE (2024,12,31),
            DAY
        )
    ) > 90
)

```

Purpose: Customers inactive for more than 90 days.

Churn Rate

```

DIVIDE (
    [Churned Customers],
    [Total Customers],
    0
) * 100

```

Purpose: Percentage of lost customers.

High Risk Customers

```

CALCULATE (
    DISTINCTCOUNT (ecommerce_orders[customer_id]),
    FILTER (
        VALUES (ecommerce_orders[customer_id]),
        CALCULATE (
            DATEDIFF (
                MAX (ecommerce_orders[order_date]),
                DATE (2024,12,31),
                DAY
            )
        ) >= 60
        &&
        CALCULATE (
            DATEDIFF (
                MAX (ecommerce_orders[order_date]),
                DATE (2024,12,31),
                DAY
            )
        ) <= 90
    )
)

```

)

Purpose: Customers likely to churn soon.

Active Customers

```
CALCULATE (
    DISTINCTCOUNT (ecommerce_orders[customer_id]),
    FILTER (
        VALUES (ecommerce_orders[customer_id]),
        CALCULATE (
            DATEDIFF (
                MAX (ecommerce_orders[order_date]),
                DATE (2024, 12, 31),
                DAY
            )
        ) <= 90
    )
)
```

Purpose: Recently active customers.

RFM Analysis

Champions

```
CALCULATE (
    DISTINCTCOUNT (ecommerce_orders[customer_id]),
    FILTER (
        VALUES (ecommerce_orders[customer_id]),
        CALCULATE (
            DATEDIFF (
                MAX (ecommerce_orders[order_date]),
                DATE (2024, 12, 31),
                DAY
            )
        ) <= 30
    )
    &&
    CALCULATE (
        DISTINCTCOUNT (ecommerce_orders[order_id])
    )
)
```

```

    ) >= 5
  )
)

```

Purpose: Most valuable customers.

At Risk Customers

```

CALCULATE (
    DISTINCTCOUNT (ecommerce_orders [customer_id]),
    FILTER (
        VALUES (ecommerce_orders [customer_id]),
        CALCULATE (
            DATEDIFF (
                MAX (ecommerce_orders [order_date]),
                DATE (2024, 12, 31),
                DAY
            )
        ) >= 60
    ) &&
    CALCULATE (
        DATEDIFF (
            MAX (ecommerce_orders [order_date]),
            DATE (2024, 12, 31),
            DAY
        )
    ) <= 120
    ) &&
    CALCULATE (
        DISTINCTCOUNT (ecommerce_orders [order_id])
    ) >= 2
)
)

```

Purpose: Previously loyal customers showing churn risk.

Lost Customers

```

CALCULATE (
    DISTINCTCOUNT (ecommerce_orders [customer_id]),
    FILTER (
        VALUES (ecommerce_orders [customer_id]),
        CALCULATE (

```

```

        DATEDIFF (
            MAX (ecommerce_orders[order_date]),
            DATE (2024,12,31),
            DAY
        )
    ) > 180
)

```

Purpose: Customers inactive for more than 180 days.

Champions Revenue %

```

DIVIDE (
    CALCULATE (
        SUM (ecommerce_orders[revenue]),
        FILTER (
            VALUES (ecommerce_orders[customer_id]),
            CALCULATE (
                DISTINCTCOUNT (ecommerce_orders[order_id])
            ) >= 5
        )
    ),
    [Total Revenue],
    0
)

```

Purpose: Revenue contribution from Champion customers.

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Tools Used: Excel | MySQL | Power BI | DAX

Project: Ecommerce Customer Retention & Churn Analysis